

Cooking Buddy

Full System Architecture

A. Entry & Mode

- User opens the website, site greets user with a welcome message (spoken)
- Site starts in voice chat mode by default
- User can switch from voice to text mode anytime

B. Login & Security

- User logs in or signs up if new
- Site checks user identity and keeps session safe

C. Pre-Cooking Info Gathering

- Site asks how many people will eat
- Site asks about diet needs, goals, and allergies
- Site shows prep time and total calories before cooking starts
- Site asks if user has a scale or measuring cups
- If no scale, site checks if a cup/volume measurement will work instead
- Site gives recipe amounts adjusted to the tool the user has (using saved conversion values)

D. Ingredient Check

- Site asks user to confirm they have all ingredients
- If something is missing, site suggests a substitute, or adds it to the shopping list
- (secondary) User uploads a photo; site identifies ingredients using Gemini's vision feature

- (secondary) Site creates a shopping list

E. Cooking

- Site guides user through steps one by one, waiting for "next"
- (secondary) User can say "next step" or "repeat" by voice, using Gemini
- Site asks about decoration/plating
- Site asks about dessert/sides
- Site asks about serving

F. Session Wrap-up

- One full "cooking session" = everything from C to E, start to finish

G. Website-Level Features (outside a cooking session)

- Site sends notifications and reminders (meal planning, shopping list, cook time)
- User can change the website's look/theme
- User can change the assistant's voice

H. Recipe Knowledge (background, not live) • Site collects recipes from Sri

Lankan recipe sites and cooking channels

- Recipes are tagged by cuisine (Sinhala / Tamil / Muslim) and linked to their source
- This happens automatically in the background, not something the user starts

Group Actions Into layers

Frontend (Client)

- 1 Voice chat screen
- 2 Text chat screen
- 3 Mic button
- 4 Photo upload button
- 5 Login/signup screen
6. Step-by-step cards
7. previous cooking sessions cards
- 8 Alerts
- 9 Settings screen (theme + voice)

Backend (Server)

- 1 Manages full session (greeting → questions → recipe → steps → end)
- 2 Login/session check
- 3 Picks the recipe
- 4 Converts measurements
- 5 Checks ingredients
- 6 Manages shopping list

External Services

- 1 Gemini API : chat & recipe suggestions
- 2 Gemini cloud API : voice (speech-to-text / text-to-speech)
- 3 Gemini API : photo-to-ingredient recognition

Database

- 1 Users
- 2 Recipes (with cuisine tag)
- 3 Ingredients
- 4 Cooking sessions
- 5 Shopping lists

Ingestion

- 1 Pulls recipes from Sri Lankan sites/channels
- 2 Tags recipes by cuisine
- 3 Saves recipes into database

System Workflows

Flow 1 – Login / Signup

- User enters email and password.
- Frontend sends credentials.
- Backend verifies the user and generates a session token.
- Backend returns the token and user profile.
- Frontend stores the token and redirects it to the Chat screen.

Flow 2 – Welcome, Diet & Allergy Gathering

- User opens the application.
- Frontend starts a session.
- Backend loads the user's dietary preferences and allergies.
- Gemini confirms or updates serving size, diet, and allergies.
- Backend saves updated information.
- Frontend displays and speaks the confirmation using Gemini TTS.

Flow 3 – Recipe Recommendation

- Users provide cooking goals, available time, and cuisine preference.
- Frontend sends the request.
- Backend adds the user's dietary preferences, allergies, and serving size.
- Gemini generates a recipe recommendation.
- Backend saves the cooking session.
- Frontend displays the recipe, calories, and preparation time.

Flow 4 – Measurement Tool Check

- Frontend asks whether to use a scale or measuring cups.
- User selects a measurement type.
- Frontend sends the selection.
- Backend retrieves conversion factors and updates the session.
- Frontend displays the recipe with converted measurements.

Flow 5 – Ingredient Check & Missing Items

- Frontend displays the ingredient list.
- User marks missing ingredients.
- Frontend sends the list.
- Backend checks for ingredient substitutes.
- If a substitute exists, it is suggested.
- Otherwise, the ingredient is added to the shopping list.
- Frontend displays the result.

Flow 6 – Photo Upload & Ingredient Recognition

- User uploads or captures a photo of ingredients.
- Frontend sends the image.
- Backend forwards the image to Gemini Vision.
- Gemini identifies the ingredients.
- Backend matches them with the ingredient database.
- Frontend displays the recognized ingredients for confirmation.

Flow 7 – Step-by-Step Cooking

- Backend provides the current cooking step.
- User says "**Next**" or taps the **Next** button.
- Frontend sends the request.
- Backend updates the current cooking step.
- Gemini converts the instruction to speech (voice mode).
- Frontend displays or plays the next instruction.

Flow 8 – Decoration, Dessert & Serving Suggestions

- Backend detects that cooking is complete.
- Gemini generates decoration, dessert, and serving suggestions.
- Backend returns the suggestions.
- Frontend displays them and marks the session as completed.

Flow 9 – Notifications & Reminders

- Backend scheduler checks shopping lists and meal plans.
- Backend creates notifications.
- Push Notification Service sends reminders at the scheduled time.
- Frontend displays the notification.

Flow 10 – Theme & Voice Customization

- User changes the theme or voice in Settings.
- Frontend sends the preferences (POST /preferences).
- Backend saves the user preferences.
- Frontend updates the UI theme and Gemini voice.

Flow 11 – Sri Lankan Recipe Ingestion (Offline)

- Scheduled pipeline collects recipes from Sri Lankan websites and cooking channels.
- Gemini structures the extracted recipe data.
- Pipeline stores recipes and ingredients in the database.
- New recipes become available for future recommendations.

Full Diagram

